

Q1

Plane is $x + y - 2z = 3$

$$\Rightarrow \frac{x-2}{1} = \frac{y-1}{1} = \frac{z-6}{-2} = \frac{-2(2+1-12-3)}{6}$$

$$\Rightarrow (x, y, z) = (6, 5, -2)$$

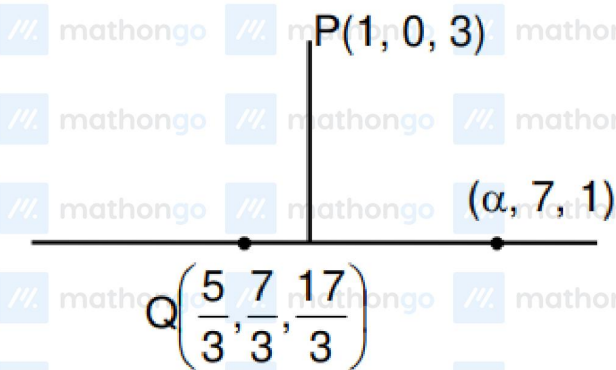
Q2

Since PQ is perpendicular to L , therefore

$$\left(1 - \frac{5}{3}\right) \left(\alpha - \frac{5}{3}\right) + \left(\frac{-7}{3}\right) \left(7 - \frac{7}{3}\right) + \left(3 - \frac{17}{3}\right) \left(1 - \frac{17}{3}\right) = 0$$

$$\Rightarrow \frac{-2\alpha}{3} + \frac{10}{9} - \frac{98}{9} + \frac{112}{9} = 0$$

$$\Rightarrow \frac{2\alpha}{3} = \frac{24}{9} \Rightarrow \alpha = 4$$



Q3

$$\vec{w} = \hat{i} + \hat{j} - 3\hat{k}$$

$$\vec{p} = 3\hat{i} - \hat{j} + \hat{k}$$

$$\vec{q} = -3\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 1 \\ -3 & 2 & 4 \end{vmatrix} = -6\hat{i} - 15\hat{j} + 3\hat{k}$$

$$\text{S.D} = \frac{|\vec{AB} \cdot (\vec{p} \times \vec{q})|}{|\vec{p} \times \vec{q}|} = \frac{|36 + 225 + 9|}{\sqrt{36 + 225 + 9}} = 3\sqrt{30}$$

Q4

d.r of normal to the plane

$$\frac{10}{3}, \frac{10}{3}, \frac{10}{3}$$

$$1, 1, 1$$

midpoint of P and Q is $\left(-\frac{2}{3}, \frac{1}{3}, \frac{4}{3}\right)$ equation of plane $x + y + z = 1$

Q5

$$\Delta = 0 \Rightarrow \begin{vmatrix} 1 & 4 & -2 \\ 1 & 7 & -5 \\ 1 & 5 & \alpha \end{vmatrix} = 0$$

$$(7\alpha + 25) - (4\alpha + 10) + (-20 + 14) = 0$$

$$3\alpha + 9 = 0 \Rightarrow \alpha = -3$$

$$\text{Also } D_z = 0 \Rightarrow \begin{vmatrix} 1 & 4 & 1 \\ 1 & 7 & \beta \\ 1 & 5 & 5 \end{vmatrix} = 0$$

$$1(35 - 5\beta) - (15) + 1(4\beta - 7) = 0$$

$$\beta = 13$$

Q6

Lines must be intersecting

$$\Rightarrow (2s - 1, 4s + 3, 3s - 1) = (2t - 3, 6t - 2, \lambda t + 1)$$

$$2s - 1 = 2t - 3, 4s + 3 = 6t - 2, 3s - 1 = \lambda t + 1$$

$$\Rightarrow t = \frac{1}{2}, s = -\frac{1}{2}, \lambda = -7$$

distance of plane contains given lines from given

plane is same as distance between point $(-3, -2, 1)$

from given plane.

Required distance equal to

$$\frac{|-69 + 20 - 2 + 48|}{\sqrt{529 + 100 + 4}} = \frac{3}{\sqrt{633}} = \frac{k}{\sqrt{633}} \Rightarrow k = 3$$

Q7

$$\vec{b} \cdot \vec{c} = 10 \Rightarrow |\vec{b}| |\vec{c}| \cos\left(\frac{\pi}{3}\right) = 10 \Rightarrow 5 \cdot |\vec{c}| \cdot \frac{1}{2} = 10$$

$$\Rightarrow |\vec{c}| = 4$$

$$\text{Also } \vec{a} \cdot (\vec{b} \times \vec{c}) = 0$$

$$|\vec{a} \times (\vec{b} \times \vec{c})| = |\vec{a}| |\vec{b} \times \vec{c}| \sin\left(\frac{\pi}{2}\right)$$

$$\sqrt{3} \times |\vec{b}| |\vec{c}| \sin \frac{\pi}{3} \times 1 = \sqrt{3} \times 5 \times 4 \times \frac{\sqrt{3}}{2} = 30$$

Q8

Let the plane be $Ax + By + Cz + 1 = 0$

Satisfying the conditions we get

$$A + 2B + C + 1 = 0 \dots (1) \text{ (passes through } (1,2,1))$$

$$2A + B + 2C + 1 = 0 \dots (2) \text{ (passes through } (2,1,2))$$

$$\frac{A}{2} + \frac{B}{3} = 0 \dots (3) \text{ (parallel to } 2x = 3y, z = 1)$$

$$\text{Solving we get } A = \frac{2}{9}, B = -\frac{1}{3} \text{ \& } C = -\frac{5}{9}$$

$$\text{Hence plane is } 2x - 3y - 5z + 9 = 0$$

$\therefore$ Passes through $(-2,0,1)$

Q9

The equation of plane passing through $(3,1,1)$ is-

$$a(x-3) + b(y-1) + c(z-1) = 0$$

$\therefore$ This plane contains the lines having drs

$$(1, -2, 2) \text{ \& } (2, 3, -1)$$

s0,

$$a - 2b + 2c = 0$$

$$2a + 3b - c = 0$$

$$\frac{a}{-2} = \frac{b}{1} = \frac{c}{1-2} \Rightarrow \frac{a}{2-6} = \frac{b}{-(-1-4)} = \frac{c}{3+4}$$

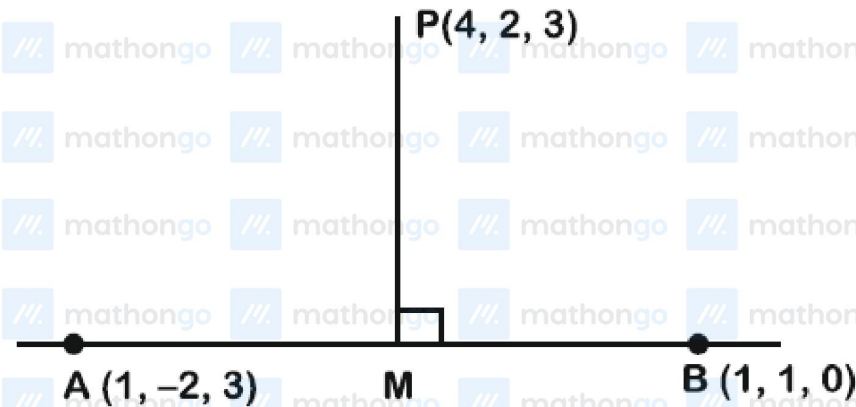
$$\frac{3}{3} = \frac{-1}{-1} = \frac{2}{2} = \frac{-1}{-1} = \frac{2}{2} = \frac{3}{3} \Rightarrow \frac{a}{4} = \frac{b}{5} = \frac{c}{7}$$

So, equation of plane is

$$-4(x-3) + 5(y-1) + 7(z-1) = 0$$

Q10

Equation of line through points $(1, -2, 3)$ and $(1, 1, 0)$ is



$$\frac{x-1}{0} = \frac{y+2}{-3} = \frac{z-3}{-3} (= \lambda \text{ say})$$

A point on above line $M = (1, -\lambda + 1, \lambda)$

$$\vec{D} \cdot \vec{R}^s \text{ of } PM = \langle -3, -\lambda - 1, \lambda - 3 \rangle$$

$\therefore PM \perp AB$

$$(-3) \cdot 0 + (-1 - \lambda)(-1) + (\lambda - 3) \cdot 1 = 0$$

$$\therefore \lambda = 1$$

$\therefore$ foot of perpendicular $= (1, 0, 1)$

This point lies on plane $2x + y - z = 1$

Q11

$$L_1 \equiv \vec{r} = (\hat{i} - \hat{j}) + 1(2\hat{i} + \hat{k})$$

$$L_2 \equiv \vec{r} = (2\hat{i} - \hat{j}) + m(\hat{i} + \hat{j} - \hat{k})$$

Equating coeff. of $\hat{i}$, $\hat{j}$ and $\hat{k}$ of L_1 and L_2

$$2 + 1 = m + 2 \quad \therefore (1)$$

$$-1 = -1 + m$$

$$1 = m$$

$\Rightarrow m = 1 = 0$ Which gives absurd result hence lines are skew (do not intersect) for any value of l and m .

Q12

Direction ratios of normal to plane are $\langle 2, -6, 4 \rangle$

Also plane passes through $(3, 1, 1)$

$$\therefore \text{Equation of plane } 2(x - 3) - 6(y - 1) + 4(z - 1) = 0$$

$$x - 3y + 2z = 2$$

Clearly it passes through $(4, 0, 1)$

Q13

Let $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} - \hat{k}$

$$\vec{a} \times \vec{b} = -\hat{i} + \hat{j} + \hat{k}$$

Equation of plane is $-x + y + z + d = 0$

It passes through (1,0,0)

$$\therefore \text{Plane is } x - y - z - 1 = 0$$

Foot of perpendicular

$$\frac{x-1}{1} = \frac{y-0}{-1} = \frac{z-1}{-1} = \frac{-(1-1-1)}{1+1+1}$$

$$\therefore (x, y, z) = \left(\frac{4}{3}, -\frac{1}{3}, \frac{2}{3}\right)$$

$$3(\alpha + \beta + \gamma) = 3\left(\frac{4}{3} - \frac{1}{3} + \frac{2}{3}\right) = 5$$

Q14

Equation of plane P is

$$(x + 4y - z + 7) + \lambda(3x + y + 5z - 8) = 0$$

$$\Rightarrow x(1 + 3\lambda) + y(4 + \lambda) + z(-1 + 5\lambda) + (7 - 8\lambda) = 0$$

$$\frac{1+3\lambda}{a} = \frac{4+\lambda}{b} = \frac{5\lambda-1}{6} = \frac{7-8\lambda}{-15}$$

From last two : $\lambda = -1$

$$\frac{-2}{a} = \frac{3}{b} = -1$$

$$\therefore a = 2, b = -3$$

Plane is $2x - 3y + 6z - 15 = 0$

$$\text{Distance} = \frac{|6-6-6-15|}{7} = \frac{21}{7} = 3$$

Q15

Equation of line through point (1,-2,3) and

parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6} = (\lambda \text{ Say})$$

A point on whole line = $(2\lambda + 1, 3\lambda - 2, -6\lambda + 3)$

This point lies on plane $x - y + 2 = 5$

$$\therefore 2\lambda + 1 - 3\lambda + 2 - 6\lambda + 3 = 5$$

$$\therefore \lambda = \frac{1}{7}$$

$\therefore$ Point on plane

$$= \left(\frac{2}{7} + 1, \frac{3}{7} - 2, -\frac{6}{7} + 3\right) = \left(\frac{9}{7}, \frac{11}{7}, \frac{15}{7}\right)$$

$\therefore$ Required distance

$$= \sqrt{\left(\frac{9}{7} - 1\right)^2 + \left(-\frac{11}{7} + 2\right)^2 + \left(\frac{15}{7} - 3\right)^2}$$

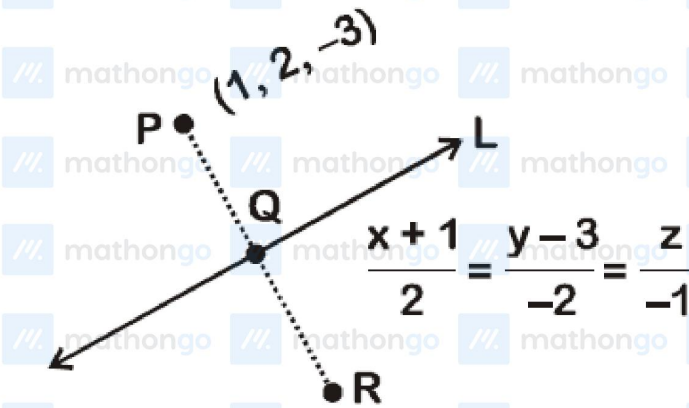
$$= 1$$

Q16

Equation of line : $\frac{x+1}{2} = \frac{y-3}{-2} = \frac{z}{-1}$ ($= \lambda$ say)

a point on line L is

$$= Q(2\lambda - 1, -2\lambda + 3, -\lambda)$$



$$D \cdot R^s \text{ of } PQ = \langle 2\lambda - 2, -2\lambda + 1, -\lambda + 3 \rangle$$

$\therefore PQ$ is perpendicular to line L

$$\therefore 2(2\lambda - 2) - 2(-2\lambda + 1) - 1(-\lambda + 3) = 0$$

$$4\lambda - 4 + 4\lambda - 2 + \lambda - 3 = 0$$

$$9\lambda - 9 = 0$$

$$\Rightarrow \lambda = 1$$

$\therefore$ Coordinate of foot of $\perp = Q = (1, 1, -1)$

$\therefore$ Coordinate of image $R = (1, 0, 1) = (a, b, c)$

$$\therefore a + b + c = 2$$

Q17

The lines are $L_1 : \frac{x+1}{2} = \frac{y-2}{-1} = \frac{z-1}{1}$

and $L_2 : \frac{x+2}{\alpha} = \frac{y+1}{5-\alpha} = \frac{z+1}{1}$ are coplanar

$$\begin{vmatrix} 1 & 3 & 2 \\ 2 & -1 & 1 \\ \alpha & 5-\alpha & 1 \end{vmatrix} = 0$$

$$1(-1-5+\alpha) - 3(2-\alpha) + 2(10-2\alpha+\alpha) = 0$$

$$\therefore \alpha = -4$$

$$\therefore \text{Equation of } L_2 : \frac{x+2}{-4} = \frac{y+1}{9} = \frac{z+1}{1}$$

$\therefore$ Point (2,-10,-2) lies on line L_2

Q18

First we will find the equation of line $x + y + z + 1 = 0 = 2x - y + z + 3$ in symmetrical form.

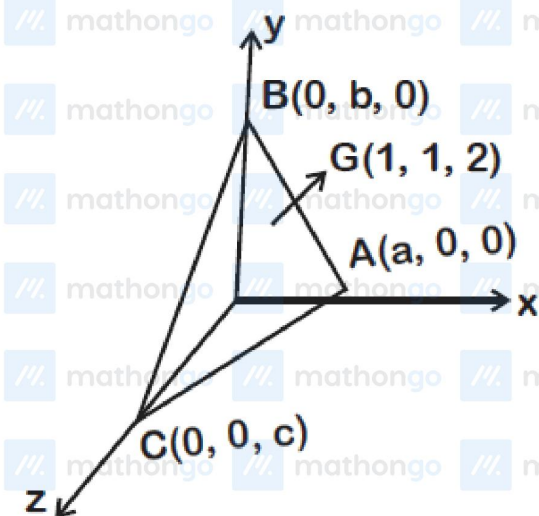
$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = 2\hat{i} + \hat{j} - 3\hat{k}$$

$$L_1 : \frac{x+2}{2} = \frac{y}{1} = \frac{z-1}{-3} \equiv \bar{r} = (-2\hat{i} + \hat{k}) + \lambda(2\hat{i} + \hat{j} - 3\hat{k})$$

$$\text{and } L_2 : \bar{r} = (\hat{i} - \hat{j}) + \mu(-\hat{j} + \hat{k}) \text{ here } \bar{\mathbf{b}}_1 \times \bar{\mathbf{b}}_2 = -2[\hat{i} + \hat{j} + \hat{k}]$$

$$\text{Shortest distance} = \frac{1}{\sqrt{3}}$$

Q19



$a = 3, b = 3$ and $c = 6$ as G is centroid

$$\vec{AB} = 3\hat{j} - 3\hat{i} \text{ and } \vec{AC} = 6\hat{k} - 3\hat{i}$$

$$\vec{AB} \times \vec{AC} = 9(2\hat{i} + 2\hat{j} + \hat{k})$$

$$\text{Required line } \frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{1}$$

Q1 JEE Main 2020 - 7 Jan (Morning)

Let P be a plane passing through the points $(2, 1, 0)$, $(4, 1, 1)$ and $(5, 0, 1)$ and R be any point $(2, 1, 6)$. Then the image of R in the plane P is:

- (A) $(6, 5, -2)$
- (B) $(6, -5, 2)$
- (C) $(2, -3, 4)$
- (D) $(2, -5, 6)$

Q2 JEE Main 2020 - 7 Jan (Evening)

If the foot of the perpendicular drawn from the point $(1, 0, 3)$ on a line passing through $(\alpha, 7, 1)$ is $\left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3}\right)$, then α is equal to

Q3 JEE Main 2020 - 8 Jan (Morning)

The shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-8}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is :

- (A) $3\sqrt{30}$
- (B) $2\sqrt{30}$
- (C) $\sqrt{30}$
- (D) $4\sqrt{30}$

Q4 JEE Main 2020 - 8 Jan (Evening)

The mirror image of the point $(1, 2, 3)$ in a plane is $\left(-\frac{7}{3}, -\frac{4}{3}, -\frac{1}{3}\right)$. Which of the following points lies on this plane?

- (A) $(-1, 1, -1)$
- (B) $(-1, -1, -1)$
- (C) $(-1, -1, 1)$
- (D) $(1, -1, 1)$

Q5 JEE Main 2020 - 9 Jan (Morning)

If plane

$$x + 4y - 2z = 1$$

$$x + 7y - 5z = \beta$$

$$x + 5y + \alpha z = 5$$

intersects in a line $(R \times R \times R)$ then $\alpha + \beta$ is equal to

- (A) 0
(B) 10
(C) -10
(D) 2

Q6 JEE Main 2020 - 9 Jan (Evening)

If the distance between the plane, $23x - 10y - 2z + 48 = 0$ and the plane containing the lines

$$\frac{x+1}{2} = \frac{y-3}{4} = \frac{z+1}{3} \text{ and } \frac{x+3}{2} = \frac{y+2}{6} = \frac{z-1}{\lambda} (\lambda \in R) \text{ is equal to } \frac{k}{\sqrt{633}}, \text{ then } k \text{ equal to}$$

Q7 JEE Main 2020 - 9 Jan (Evening)

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 5$, $\vec{b} \cdot \vec{c} = 10$ and angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. If $\vec{a}$ is perpendicular to the vector $\vec{b} \times \vec{c}$, then $|\vec{a} \times (\vec{b} \times \vec{c})|$ is equal to

Q8 JEE Main 2020 - 2 Sep (Morning)

The plane passing through the points $(1, 2, 1)$, $(2, 1, 2)$ and parallel to the line, $2x = 3y, z = 1$ also passes through the point:

- (A) $(0, 6, -2)$
(B) $(-2, 0, 1)$
(C) $(0, -6, 2)$
(D) $(2, 0, -1)$

Q9 JEE Main 2020 - 2 Sep (Evening)

A plane passing through the point $(3, 1, 1)$ contains two lines whose direction ratios are 1, -2, 2 and 2, 3, -1 respectively. If this plane also passes through the point $(\alpha, -3, 5)$, then α is equal to:

- (A) 5
(B) 10
(C) -10
(D) -5

Q10 JEE Main 2020 - 3 Sep (Morning)

The foot of the perpendicular drawn from the point $(4, 2, 3)$ to the line joining the points $(1, -2, 3)$ and $(1, 1, 0)$ lies on the plane:

- (A) $x - 2y + z = 1$

(B) $x + 2y - z = 1$

(C) $x - y - 2y = 1$

(D) $2x + y - z = 1$

Q11 JEE Main 2020 - 3 Sep (Morning)

The lines

$$\vec{r} = (\hat{i} - \hat{j}) + 1(2\hat{i} + \hat{k}) \text{ and}$$

$$\vec{r} = (2\hat{i} - \hat{j}) + m(\hat{i} + \hat{j} - \hat{k})$$

(A) do not intersect for any values of 1 and m (B) intersect for all values of 1 and m (C) intersect when $1 = 2$ and $m = \frac{1}{2}$ (D) intersect when $1 = 1$ and $m = 2$ **Q12 JEE Main 2020 - 3 Sep (Evening)**

The plane which bisects the line joining, the points (4,-2,3) and (2,4,-1) at right angles also passes through the point:

(A) (4,0,1)

(B) (0,-1,1)

(C) (0,1,-1)

(D) (4,0,-1)

Q13 JEE Main 2020 - 3 Sep (Evening)Let a plane P contain two lines

$$\vec{r} = \hat{i} + \lambda(\hat{i} + \hat{j}), \lambda \in \mathbb{R} \text{ and}$$

$$\vec{r} = -\hat{j} + \mu(\hat{j} - \hat{k}), \mu \in \mathbb{R}$$

If $Q(\alpha, \beta, \gamma)$ is the foot of the perpendicular drawn from the point $M(1, 0, 1)$ to P , then $3(\alpha + \beta + \gamma)$ equals**Q14 JEE Main 2020 - 4 Sep (Morning)**

If the equation of a plane P , passing through the intersection of the planes $x + 4y - z + 7 = 0$ and $3x + y + 5z = 8$ is $ax + by + 6z = 15$ for some $a, b \in \mathbb{R}$, then the distance of the point (3,2,-1) from the plane P is

Q15 JEE Main 2020 - 4 Sep (Evening)

The distance of the point (1,2,3) from the plane $x - y + z = 5$ measured parallel to the line

$\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is :

- (A) 7
- (B) $\frac{7}{5}$
- (C) 1
- (D) $\frac{1}{7}$

Q16 JEE Main 2020 - 5 Sep (Morning)

If (a, b, c) is the image of the point (1,2,-3) in the line, $\frac{x+1}{2} = \frac{y-3}{-2} = \frac{z}{-1}$, then $a + b + c$ is equal to :

- (A) 2
- (B) 1
- (C) 3
- (D) -1

Q17 JEE Main 2020 - 5 Sep (Evening)

If for some $\alpha \in \mathbb{R}$, the lines $L_1 : \frac{x+1}{2} = \frac{y-2}{-1} = \frac{z-1}{1}$ and $L_2 : \frac{x+2}{\alpha} = \frac{y+1}{5-\alpha} = \frac{z+1}{1}$ are coplanar then the line L_2 passes through the point:

- (A) (10,2,2)
- (B) (2,-10,-2)
- (C) (10,-2,-2)
- (D) (-2,10,2)

Q18 JEE Main 2020 - 6 Sep (Morning)

The shortest distance between the lines $\frac{x-1}{0} = \frac{y+1}{-1} = \frac{z}{1}$ and $x + y + z + 1 = 0, 2x - y + z + 3 = 0$ is :

- (A) $\frac{1}{\sqrt{2}}$
- (B) 1
- (C) $\frac{1}{\sqrt{3}}$
- (D) $\frac{1}{2}$

Q19 JEE Main 2020 - 6 Sep (Evening)

A plane P meets the coordinate axes at A , B and C respectively. The centroid of $\triangle ABC$ is given to be $(1, 1, 2)$. Then the equation of the line through this centroid and perpendicular to the plane P is

(A) $\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{1}$

(B) $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-2}{2}$

(C) $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$

(D) $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-2}{1}$

Answer Key

Q1 (A)

Q2 (4)

Q3 (A)

Q4 (D)

Q5 (B)

Q6 (3)

Q7 (30)

Q8 (B)

Q9 (A)

Q10 (D)

Q11 (A)

Q12 (D)

Q13 (5)

Q14 (3)

Q15 (C)

Q16 (A)

Q17 (B)

Q18 (C)

Q19 (A)